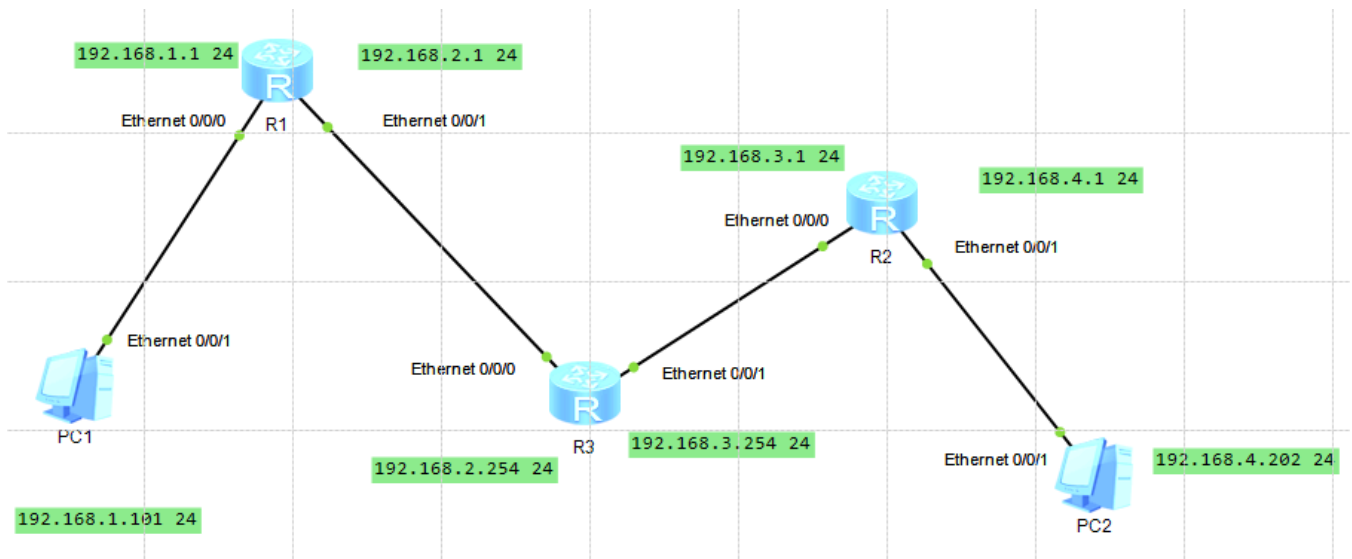


Lab 9

Practice 9.1



PC1 Configuration:

- Hostname:
- MAC address: 54-89-98-33-16-B4
- IPv4 配置: ☒ 静态, ☐ DHCP, ☐ 自动获取 DNS 服务器地址
- IP 地址: 192.168.1.101, DNS1: 0.0.0.0, DNS2: 0.0.0.0
- 子网掩码: 255.255.255.0
- 网关: 192.168.1.1
- IPv6 配置: ☒ 静态, ☐ DHCPv6
- IPv6 地址: ::, 前缀长度: 128, IPv6 网关: ::

PC2 Configuration:

- Hostname:
- MAC address: 54-89-98-3E-3D-69
- IPv4 配置: ☒ 静态, ☐ DHCP, ☐ 自动获取 DNS 服务器地址
- IP 地址: 192.168.4.202, DNS1: 0.0.0.0, DNS2: 0.0.0.0
- 子网掩码: 255.255.255.0
- 网关: 192.168.4.1
- IPv6 配置: ☒ 静态, ☐ DHCPv6
- IPv6 地址: ::, 前缀长度: 128, IPv6 网关: ::

```
[R1]display ip routing-table
Route Flags: R - relay, D - download to fib
-----
Routing Tables: Public
      Destinations : 7          Routes : 7

Destination/Mask    Proto    Pre  Cost           Flags NextHop         Interface
-----
127.0.0.0/8         Direct   0    0                D    127.0.0.1         InLoopBack0
127.0.0.1/32        Direct   0    0                D    127.0.0.1         InLoopBack0
192.168.1.0/24       Direct   0    0                D    192.168.1.1       Ethernet0/0/0
192.168.1.1/32       Direct   0    0                D    127.0.0.1         Ethernet0/0/0
192.168.2.0/24       Direct   0    0                D    192.168.2.1       Ethernet0/0/1
192.168.2.1/32       Direct   0    0                D    127.0.0.1         Ethernet0/0/1
192.168.4.0/24       Static   60    0                RD    192.168.2.254     Ethernet0/0/1

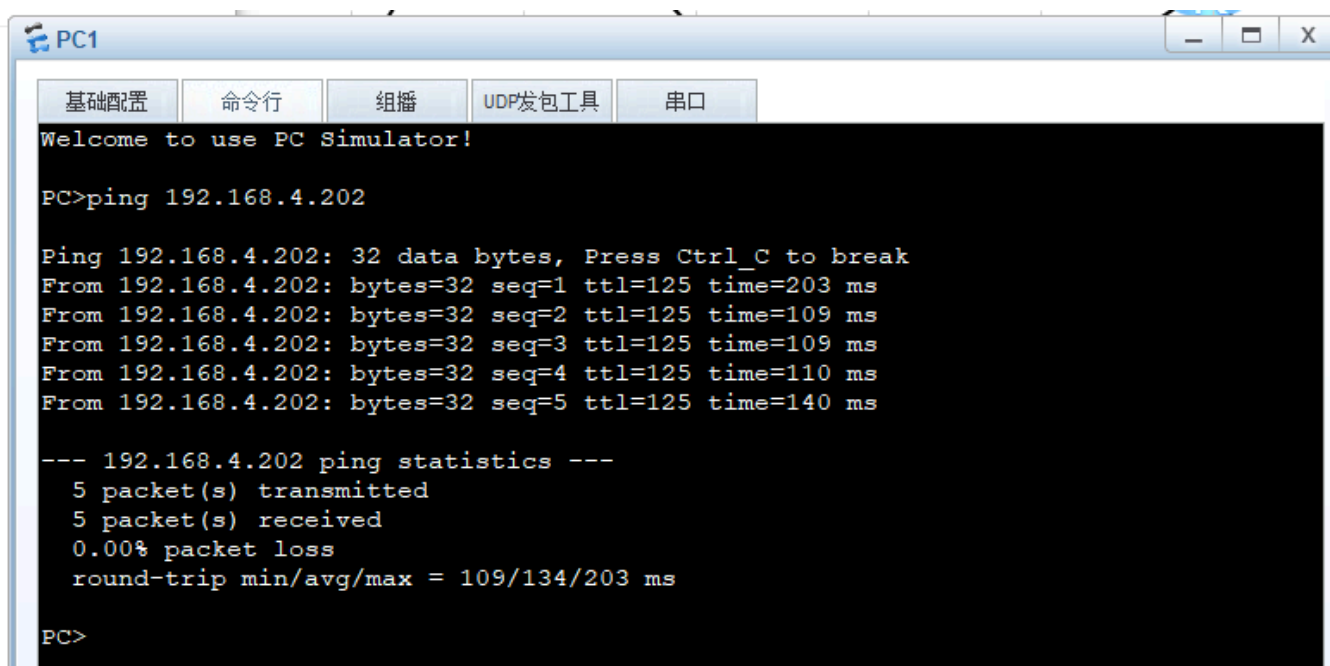
[R1]
```

```
[R2]display ip routing-table
Route Flags: R - relay, D - download to fib
-----
Routing Tables: Public
      Destinations : 7          Routes : 7

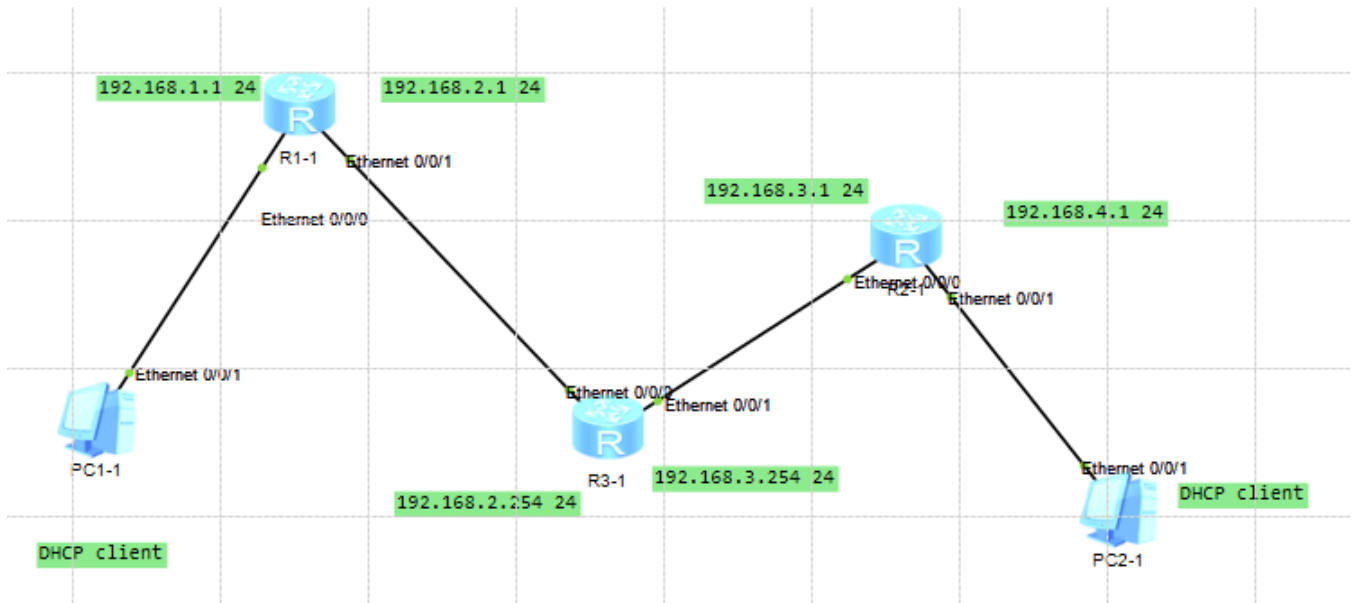
Destination/Mask    Proto   Pre  Cost           Flags NextHop         Interface
-----
 127.0.0.0/8        Direct  0    0              D   127.0.0.1         InLoopBack0
 127.0.0.1/32        Direct  0    0              D   127.0.0.1         InLoopBack0
192.168.1.0/24        Static 60    0             RD   192.168.3.254      Ethernet0/0/0
192.168.3.0/24        Direct  0    0              D   192.168.3.1        Ethernet0/0/0
192.168.3.1/32        Direct  0    0              D   127.0.0.1          Ethernet0/0/0
192.168.4.0/24        Direct  0    0              D   192.168.4.1        Ethernet0/0/1
192.168.4.1/32        Direct  0    0              D   127.0.0.1          Ethernet0/0/1
```

```
[R3]display ip routing-table
Route Flags: R - relay, D - download to fib
-----
Routing Tables: Public
      Destinations : 8          Routes : 8

Destination/Mask    Proto   Pre  Cost           Flags NextHop         Interface
-----
 127.0.0.0/8        Direct  0    0              D   127.0.0.1         InLoopBack0
 127.0.0.1/32        Direct  0    0              D   127.0.0.1         InLoopBack0
192.168.1.0/24        Static 60    0             RD   192.168.2.1        Ethernet0/0/0
192.168.2.0/24        Direct  0    0              D   192.168.2.254      Ethernet0/0/0
192.168.2.254/32      Direct  0    0              D   127.0.0.1          Ethernet0/0/0
192.168.3.0/24        Direct  0    0              D   192.168.3.254      Ethernet0/0/1
192.168.3.254/32      Direct  0    0              D   127.0.0.1          Ethernet0/0/1
192.168.4.0/24        Static 60    0             RD   192.168.3.1        Ethernet0/0/1
```



Practice 9.2



```
[R1-Ethernet0/0/0]dhcp select interface
[R1-Ethernet0/0/0]dhcp server dns-list 114.114.114.114
```

额外需要做的就是，R1开启DHCP给PC1自动分配ip，R2开启DHCP给PC2自动分配ip

Screenshots of PC configuration windows and command-line interfaces for PC1-1 and PC2-1.

PC1-1 Configuration:

- Host Name:
- MAC Address: 54-89-98-F0-40-AA
- IPv4 Configuration: ☒ DHCP, ☒ 自动获取 DNS 服务器地址
- IPv6 Configuration: ☒ 静态, ☐ DHCPv6

PC2-1 Configuration:

- Host Name:
- MAC Address: 54-89-98-F8-1F-33
- IPv4 Configuration: ☒ DHCP, ☒ 自动获取 DNS 服务器地址
- IPv6 Configuration: ☒ 静态, ☐ DHCPv6

PC1-1 Command-Line Interface:

```
PC>ipconfig
Link local IPv6 address.....: fe80::5689:98ff:fef0:40aa
IPv6 address.....: :: / 128
IPv6 gateway.....: ::
IPv4 address.....: 192.168.1.254
Subnet mask.....: 255.255.255.0
Gateway.....: 192.168.1.1
Physical address.....: 54-89-98-F0-40-AA
DNS server.....: 114.114.114.114

PC>ping 192.168.4.254

Ping 192.168.4.254: 32 data bytes, Press Ctrl_C to break
From 192.168.4.254: bytes=32 seq=1 ttl=125 time=250 ms
From 192.168.4.254: bytes=32 seq=2 ttl=125 time=109 ms
From 192.168.4.254: bytes=32 seq=3 ttl=125 time=140 ms
From 192.168.4.254: bytes=32 seq=4 ttl=125 time=125 ms
From 192.168.4.254: bytes=32 seq=5 ttl=125 time=109 ms

--- 192.168.4.254 ping statistics ---
5 packet(s) transmitted
5 packet(s) received
0.00% packet loss
round-trip min/avg/max = 109/146/250 ms
```

PC2-1 Command-Line Interface:

```
PC>ipconfig
Link local IPv6 address.....: fe80::5689:98ff:fef8:1f33
IPv6 address.....: :: / 128
IPv6 gateway.....: ::
IPv4 address.....: 192.168.4.254
Subnet mask.....: 255.255.255.0
Gateway.....: 192.168.4.1
Physical address.....: 54-89-98-F8-1F-33
DNS server.....: 114.114.114.114

PC>ping 192.168.1.254

Ping 192.168.1.254: 32 data bytes, Press Ctrl_C to break
From 192.168.1.254: bytes=32 seq=1 ttl=125 time=141 ms
From 192.168.1.254: bytes=32 seq=2 ttl=125 time=141 ms
From 192.168.1.254: bytes=32 seq=3 ttl=125 time=141 ms
From 192.168.1.254: bytes=32 seq=4 ttl=125 time=140 ms
From 192.168.1.254: bytes=32 seq=5 ttl=125 time=109 ms

--- 192.168.1.254 ping statistics ---
5 packet(s) transmitted
5 packet(s) received
0.00% packet loss
round-trip min/avg/max = 109/134/141 ms
```